

Reviewer Pack — Minimal Symplectic Leaf Order Bounds Resolution Proof Width

Entry 2aa2ca1a5738 · INCONCLUSIVE

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2026-05-29 22:29:45 UTC

Minimal Symplectic Leaf Order Bounds Resolution Proof Width

Entry ID: 2aa2ca1a5738 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 22:29:45 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Boolean Function Complexity: Tree-like Resolution Proof Complexity

Statement:

For every unsatisfiable 3-CNF F with m clauses on n variables, the minimum order of a symplectic leaf in its associated moment map is lower bounded by $\Omega(n^{1/4} * \log(m))$. Specifically, there exists a constant $c > 0$ such that for all 3-CNFs F , $\min_{\{i\}} L_i \leq c * (n^{1/4} * \log(m))$ where L_i denotes the order of symplectic leaf i in the moment map associated with F .

Rationale (proposer's reasoning):

Symplectic geometry has been previously used to study quantum complexity classes, and its structures such as leaves could potentially reveal non-trivial bounds on proof widths in resolution. The conjecture suggests that the geometric structure of symplectic leaves might provide a novel way to understand the complexity of solving SAT problems.

Taxonomy category: cg_kw_andreev (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e7a74a9b1a12b8ef

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each seed, if the mean minimum symplectic leaf order across all generated 3-CNFs is greater than or equal to $c * (n^{1/4} * \log(m))$ for a given constant c and at least 80% of seeds, then support the conjecture; otherwise falsify it.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n, m):
        clauses = []
        for _ in range(m):
            clause = [random.randint(1, n), random.randint(1, n), random.randint(1, n)]
            random.shuffle(clause)
            clauses.append(clause)
        return clauses

    def is_unsatisfiable(cnf):
        # Simple backtracking to check if the CNF is unsatisfiable
        literals = [False] * (2 * n + 1)

        def backtrack(i):
            if i == m:
                return True
            for j in range(3):
                lit = cnf[i][j]
                if not literals[lit]:
                    literals[lit] = True
                    if backtrack(i + 1):
                        return True
                    literals[lit] = False
                elif -lit in literals and literals[-lit]:
                    return False
            return False

        m = len(cnf)
        n = max(abs(lit) for clause in cnf for lit in clause)
        return not backtrack(0)
```

```

def compute_min_symplectic_leaf_order(n, m):
    # Placeholder function to simulate the computation of symplectic leaf order
    # This is a dummy implementation and should be replaced with actual computation
    return random.uniform(1, n ** 0.25 * math.log(m))

results = []
for _ in range(30): # 30 instances per seed
    n = random.randint(5, 40)
    m = random.randint(10, 100)
    cnf = generate_3cnf(n, m)
    if is_unsatisfiable(cnf):
        order = compute_min_symplectic_leaf_order(n, m)
        results.append(order)

if not results:
    return {
        "metric_name": "min_symplectic_leaf_order",
        "metric_value": 0,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

min_order = min(results)
n_max = max(n for _ in range(30))
conjecture_holds = all(order >= c * (n ** 0.25 * math.log(m)) for order, n, m in zip(results,
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "min_symplectic_leaf_order",
    "metric_value": min_order,
    "instances_tested": len(results),
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43,

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")

```

```

else:
    first_failing_seed = next(result for result in results if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\", first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ace733dd.py", line 95, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ace733dd.py", line 61, in run_trial
    if is_unsatisfiable(cnf):
        ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ace733dd.py", line 31, in is_unsatisfiable
    literals = [False] * (2 * n + 1)
                        ^
UnboundLocalError: cannot access local variable 'n' where it is not associated with a value

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating whether the mean minimum symplectic leaf order meets the support condition for the conjecture.
 | next: Investigate and fix the crash in the test code to allow for proper evaluation of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	20184
2	propose	ollama_remote	glm4:latest	0	0	13474
3	preregistration	ollama_remote	glm4:latest	0	0	18383
4	novelty	ollama_remote	glm4:latest	0	0	17439
5	novelty	ollama_remote	glm4:latest	0	0	13216
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14717
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18411
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19544
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12749

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
10	judge	ollama_remote	glm4:latest	0	0	8723

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 156839 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/2aa2ca1a5738.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/2aa2ca1a5738.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/2aa2ca1a5738.tar.gz (if generated)